

FINCASTLE AGDM, AB, Canada

WMO: 715350

Lat: 49.8023N

Lon: 112.0459W

Elev: 2638

StdP: 13.35

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.7	-12.0	-23.8	1.6	-17.4	-17.9	2.3	-12.0	30.8	40.0	27.8	34.3	6.5	280	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	27.6	89.2	62.9	85.8	62.2	82.5	60.9	66.4	81.7	64.6	80.3	62.9	78.2	8.5	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	88.6	72.7	59.0	82.1	70.7	57.0	76.4	68.5	32.5	81.7	31.1	79.9	29.7	78.3	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.7	24.3	21.1	DB	-27.1	95.9	5.5	3.1	-31.1	98.1	-34.3	99.9	-37.4	101.7	-41.4	103.9	
			WB	-27.4	70.7	5.6	2.4	-31.4	72.5	-34.7	73.9	-37.9	75.2	-42.0	76.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.1	22.2	20.7	32.4	43.0	53.4	60.6	66.7	64.5	55.9	44.0	31.7	20.6	(6)
	DBStd	19.90	15.85	15.87	13.43	9.02	7.82	5.67	5.01	5.85	8.13	9.18	13.45	15.34	(7)
	HDD50	4251	862	820	547	238	53	2	0	1	34	226	556	912	(8)
	HDD65	8193	1327	1240	1010	661	365	154	40	81	286	652	1000	1377	(9)
	CDD50	1735	0	0	3	27	160	321	518	450	212	39	5	0	(10)
	CDD65	202	0	0	0	0	6	23	93	66	14	0	0	0	(11)
	CDH74	3727	0	0	0	21	219	427	1475	1174	398	13	0	0	(12)
	CDH80	1270	0	0	0	3	46	108	546	438	128	1	0	0	(13)
	WSAvg	9.7	11.2	9.7	10.6	11.1	10.0	9.3	7.6	7.5	8.5	10.0	10.7	10.7	(14)
	PrecAvg	14.8	0.9	0.6	0.9	1.4	1.8	2.4	1.3	1.7	1.6	0.6	0.7	0.8	(15)
(16) Precipitation	PrecMax	25.6	2.7	1.1	1.8	5.8	5.3	7.1	4.7	4.8	5.2	1.9	1.7	2.2	(16)
	PrecMin	9.6	0.1	0.0	0.0	0.0	0.3	0.3	0.4	0.2	0.2	0.0	0.1	0.2	(17)
	PrecStd	4.1	0.7	0.3	0.5	1.1	1.2	1.6	1.0	1.2	1.3	0.5	0.4	0.6	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.4	57.2	64.8	77.1	85.2	88.9	94.0	92.8	89.0	75.6	64.6	53.1	(19)
		MCWB	44.4	43.0	47.7	52.4	58.4	64.2	63.4	63.1	59.7	54.0	49.6	40.7	(20)
	2%	DB	49.2	49.5	59.7	69.7	80.0	82.8	89.6	88.9	83.8	68.8	57.0	48.1	(21)
		MCWB	40.4	39.3	45.3	49.7	56.6	61.4	64.8	61.8	58.4	51.5	44.9	39.5	(22)
	5%	DB	45.2	45.1	55.7	64.4	75.3	79.0	86.0	85.0	78.7	63.6	52.8	44.3	(23)
		MCWB	37.6	37.0	42.7	47.3	54.3	59.4	64.0	60.9	57.1	49.3	43.1	37.2	(24)
	10%	DB	41.7	41.3	50.8	59.4	70.4	75.0	82.4	81.0	72.8	59.1	48.6	40.9	(25)
		MCWB	35.8	34.9	40.2	44.8	52.2	57.9	62.8	60.2	55.1	46.8	39.9	34.6	(26)
	0.4%	WB	45.4	43.8	49.2	53.7	59.9	67.0	70.3	66.7	62.8	56.2	50.5	43.0	(27)
		MCDB	54.3	54.6	61.9	74.4	81.1	82.1	84.9	83.6	79.6	71.0	63.2	50.3	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	40.8	40.1	46.1	50.6	57.2	63.9	67.5	64.6	59.9	53.0	45.9	39.5	(29)
		MCDB	49.1	48.9	58.0	68.1	77.0	78.1	82.2	81.7	78.9	66.5	55.5	46.3	(30)
	5%	WB	38.1	37.4	43.4	47.9	55.3	61.5	65.7	62.9	57.8	50.1	43.1	37.0	(31)
		MCDB	44.3	44.6	54.7	62.4	72.4	75.7	80.8	79.7	76.1	62.0	51.4	43.2	(32)
	10%	WB	36.0	34.9	40.5	45.5	53.4	59.2	64.1	61.3	55.6	47.7	40.3	34.7	(33)
		MCDB	41.3	40.8	49.9	57.7	68.3	72.0	79.4	77.1	71.8	58.0	47.6	40.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.9	20.3	21.0	22.8	23.8	22.8	27.6	27.7	25.8	22.4	19.7	18.3	(35)
		MCDBR	19.3	23.5	28.0	33.2	32.8	30.4	34.8	36.8	36.7	29.7	24.4	19.7	(36)
		MCWBR	13.0	15.0	15.7	16.5	13.7	12.9	14.4	14.2	15.0	15.3	14.5	12.9	(37)
	5% WB	MCDBR	18.8	23.2	27.2	31.0	29.4	28.1	30.8	31.8	34.2	28.0	23.3	19.5	(38)
		MCWBR	13.2	15.6	16.0	15.9	12.8	13.0	14.1	13.2	14.5	14.7	14.6	13.6	(39)
(40) Clear-Sky Solar Irradiance	taub		0.262	0.269	0.292	0.333	0.350	0.345	0.347	0.366	0.325	0.290	0.263	0.250	(40)
	taud		2.450	2.481	2.471	2.355	2.358	2.420	2.428	2.348	2.438	2.517	2.497	2.437	(41)
	Ebn at Noon		257	284	294	289	288	289	287	275	277	271	255	246	(42)
	Edh at Noon		19	23	28	36	37	35	34	36	29	22	18	16	(43)
(44) All-Sky Solar Radiation	RadAvg		395	711	1122	1550	1842	2034	2223	1778	1264	786	439	309	(44)
	RadStd		34	58	86	106	142	133	113	111	105	80	33	28	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (65 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page